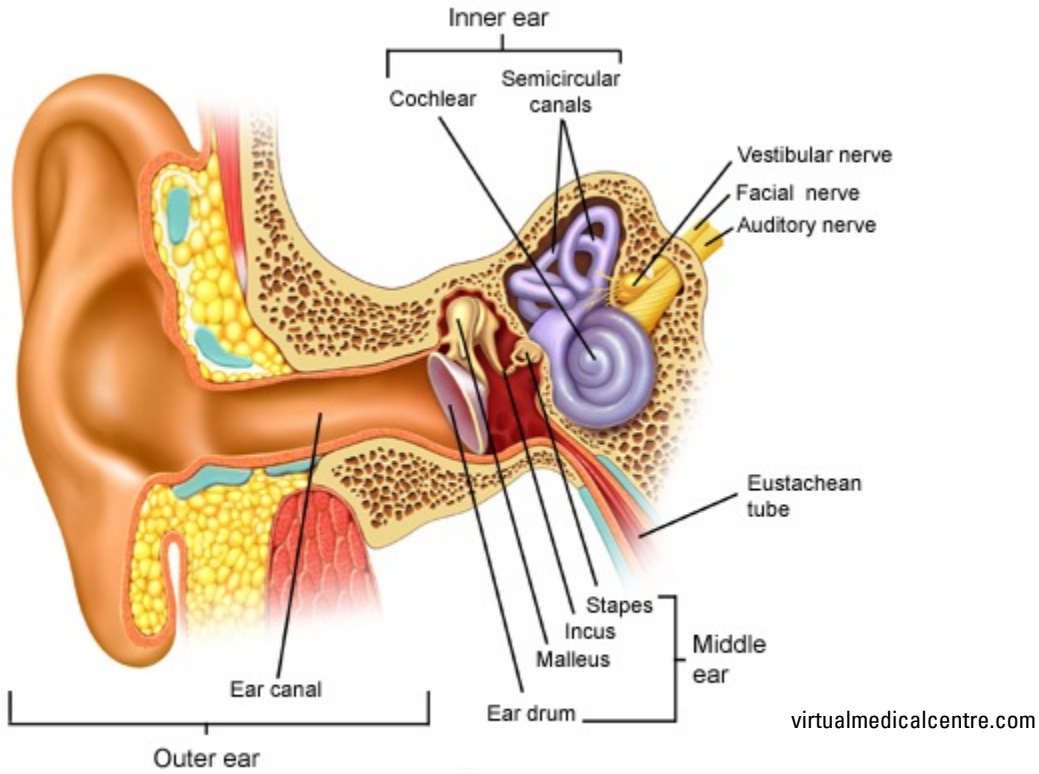
# SPATIAL DISORIENTATION

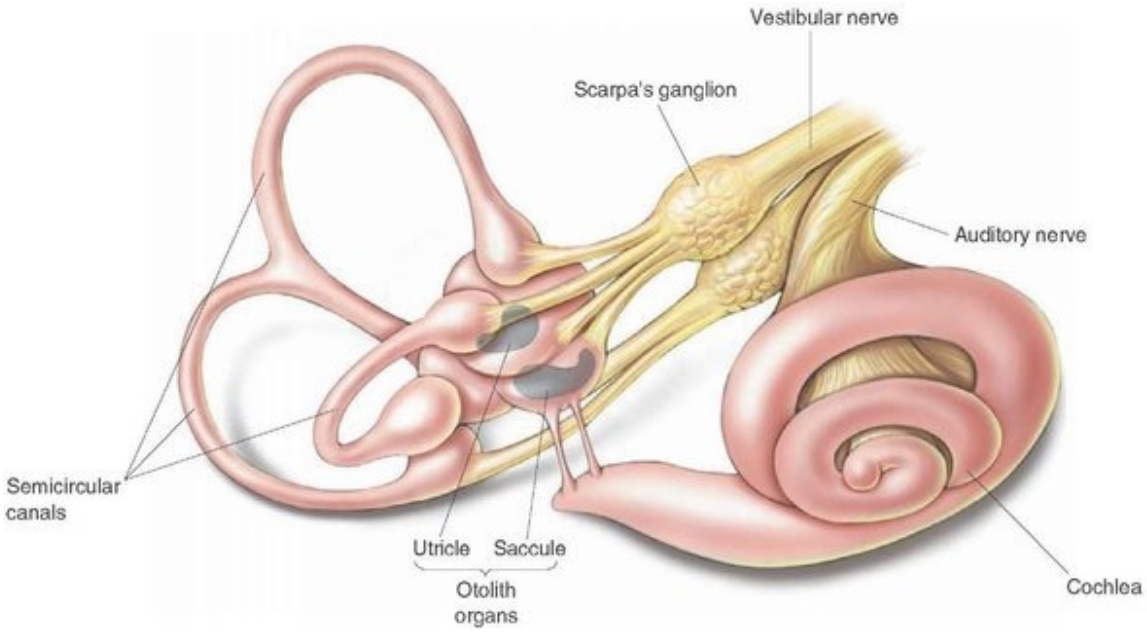
Dr. Andrew Liu

ENP-0112

# ANATOMY OF THE EAR

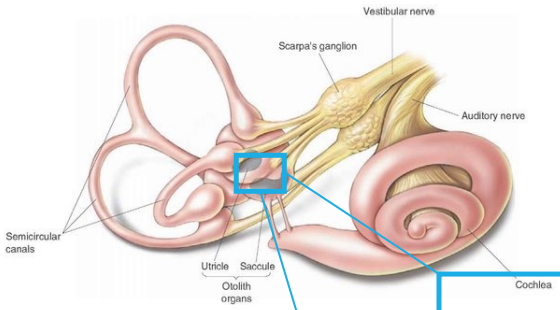


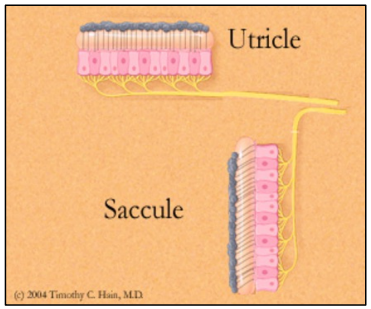
# VESTIBULAR SYSTEM



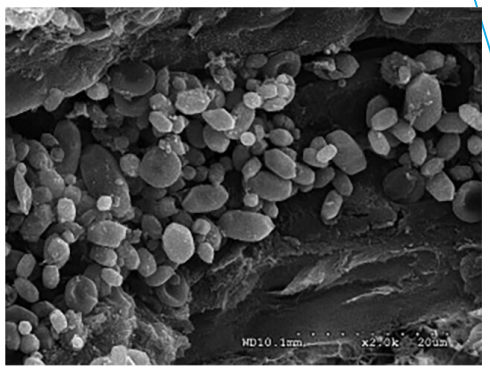

http://utahhearingandbalance.com/dizziness/balance-and-the-vestibular-system/

# OTOLITH ORGANS

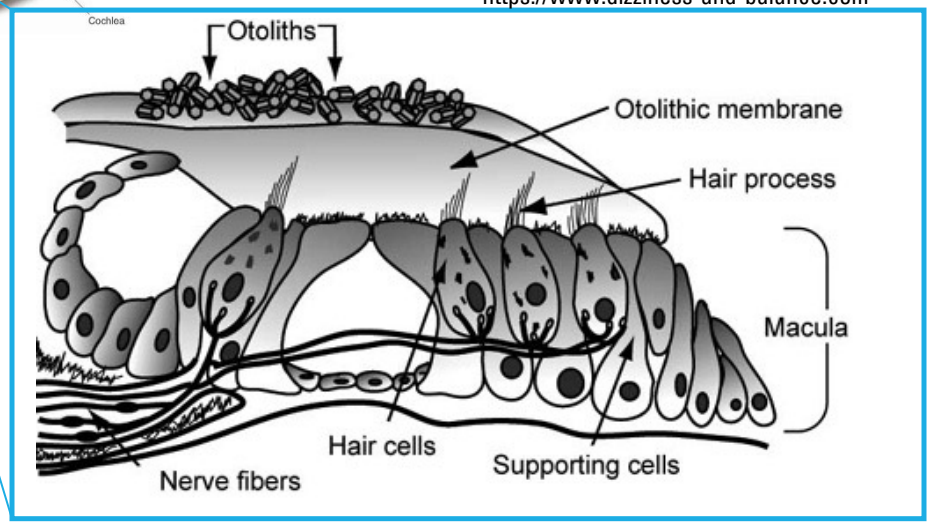



https://www.dizziness-and-balance.com

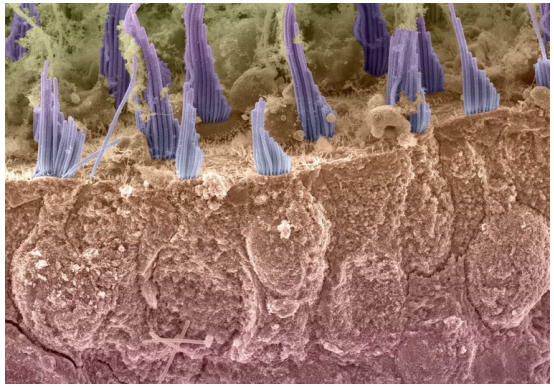


Credit: You, Instrum, and Parnes, 2019.

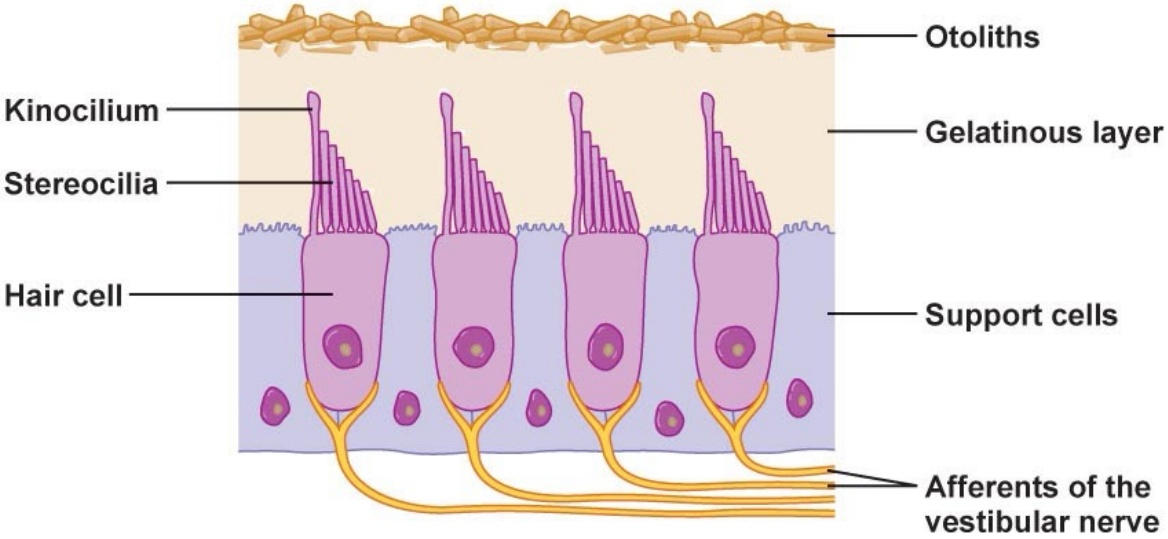



http://nasa.gov

# OTOLITH HAIR CELLS

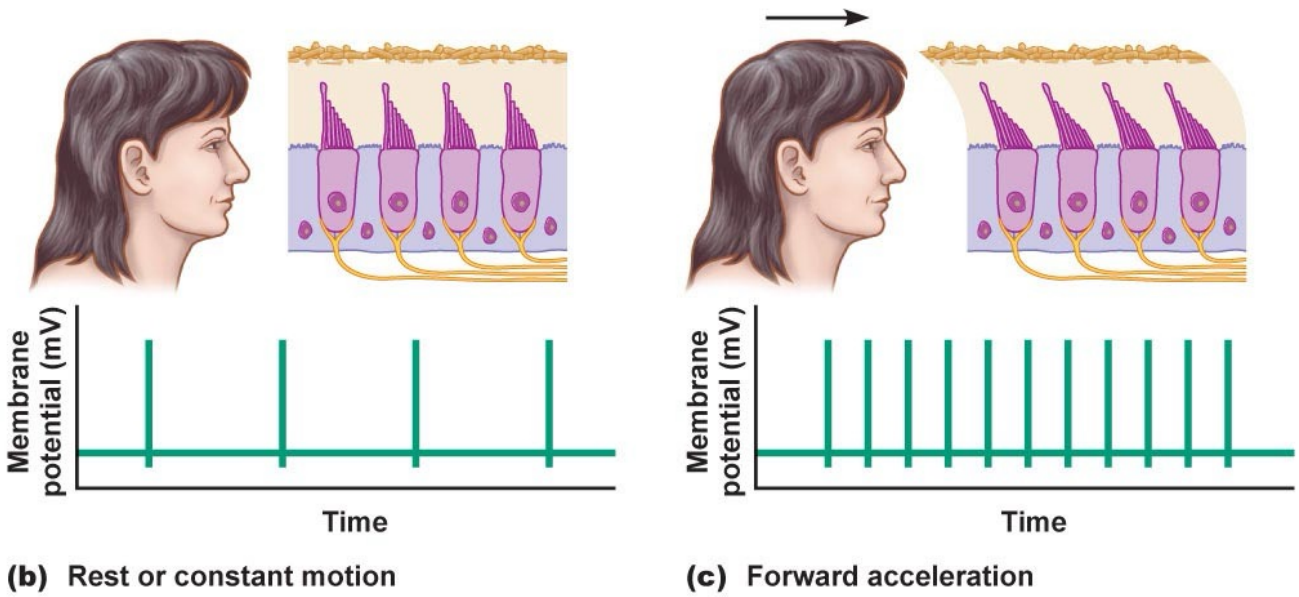


WellcomeCollection.org



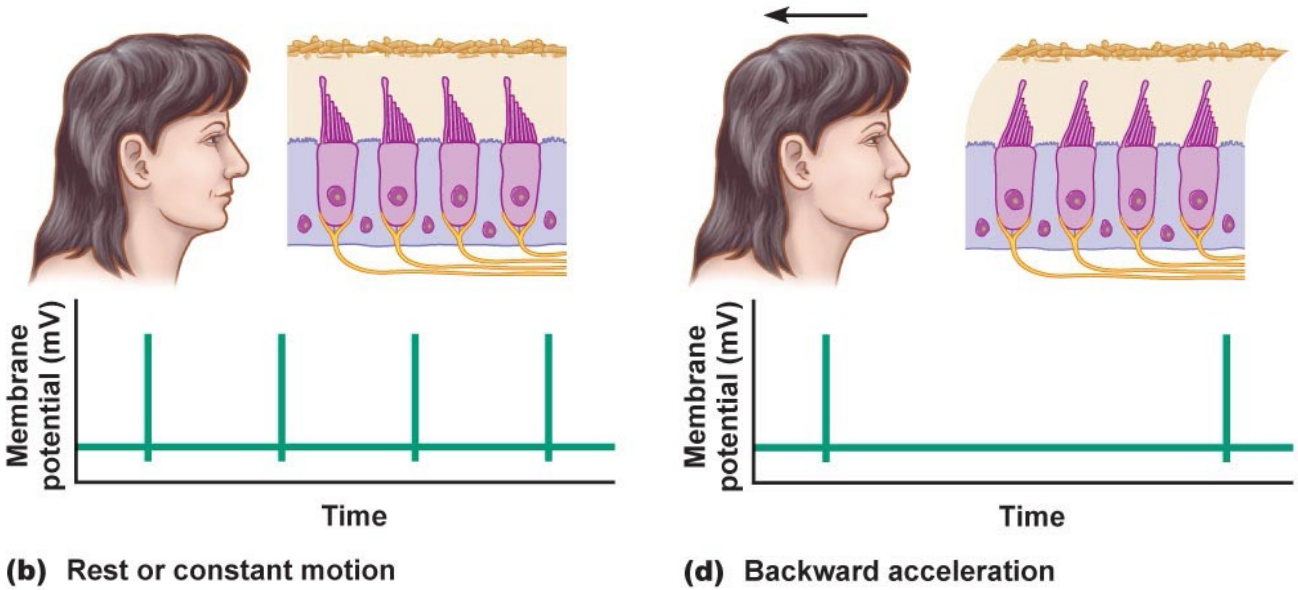

Stanfield, 2011

# DETECTING FORWARD ACCELERATION



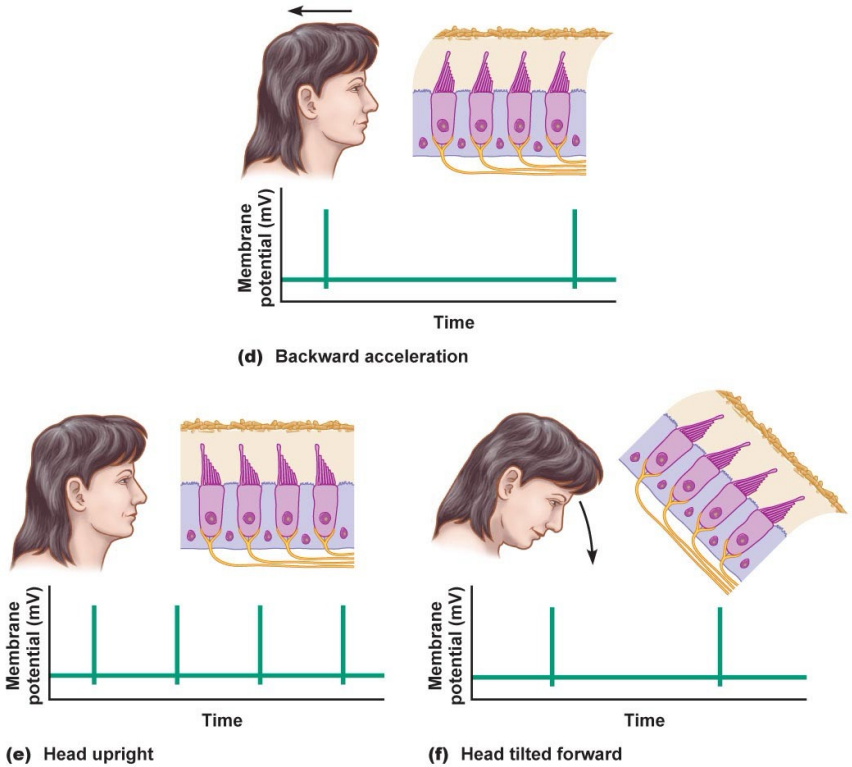

Stanfield, 2011

# DETECTING BACKWARD ACCELERATION




Stanfield, 2011

# DETECTION OF ACCELERATION V. TILT?




Stanfield, 2011

# NET GRAVITO-INERTIAL FORCE (GIF)

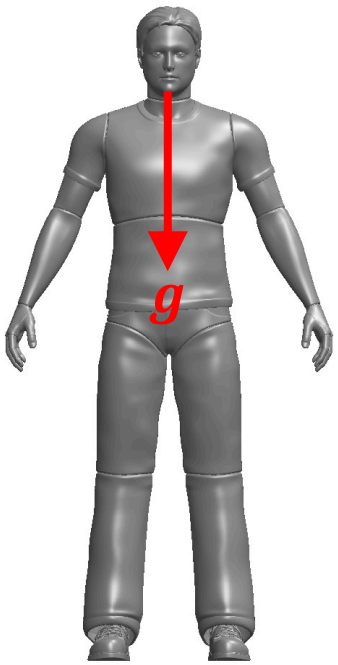
*Resultant force when combining the force due to gravity with the force due to an acceleration*

Definitions:

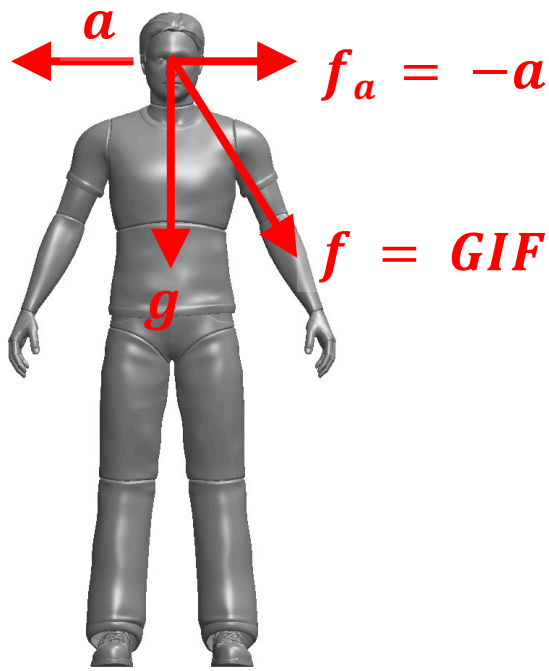
- $f$ is the gravito-inertial force per unit mass
- $a$ is the linear acceleration
- $g$ is the gravitational force per unit mass

# NET GRAVITO-INERTIAL FORCE (GIF)

Static



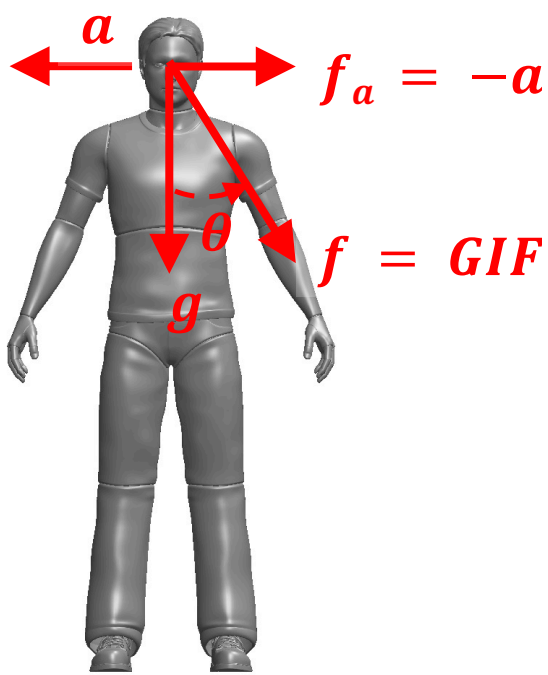

Linear Acceleration




$$f = g + f_a$$

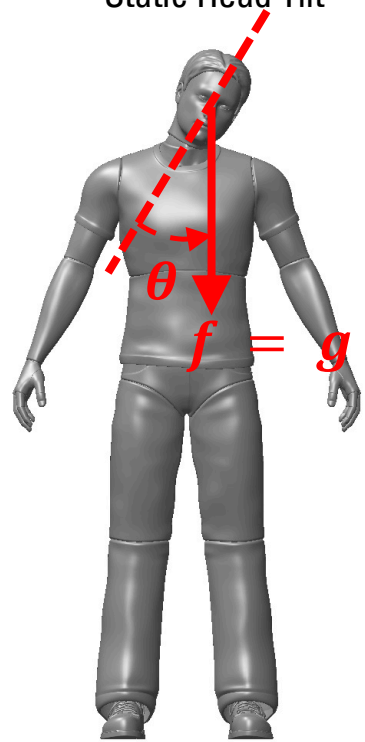
# AMBIGUITY IN NET GRAVITO-INERTIAL FORCE (GIF)

Linear Acceleration




$$f = g + f_a$$

Static Head Tilt

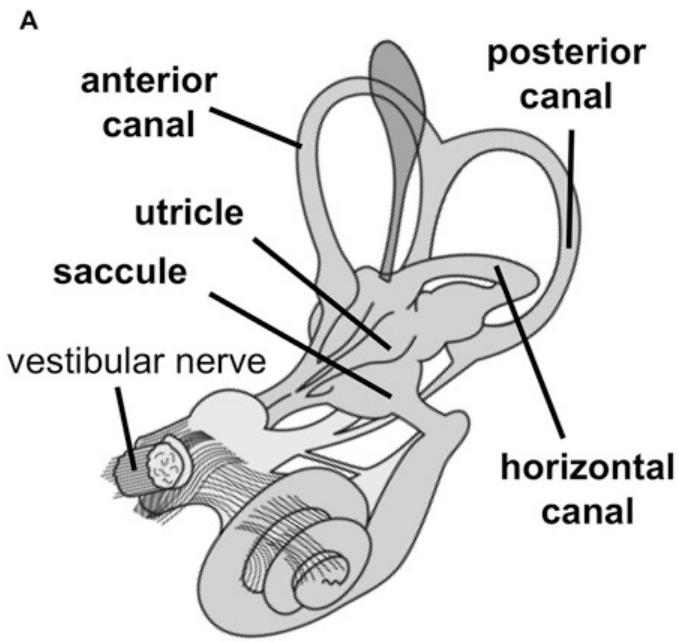


# RESOLVING AMBIGUOUS OTOLITH CUES?

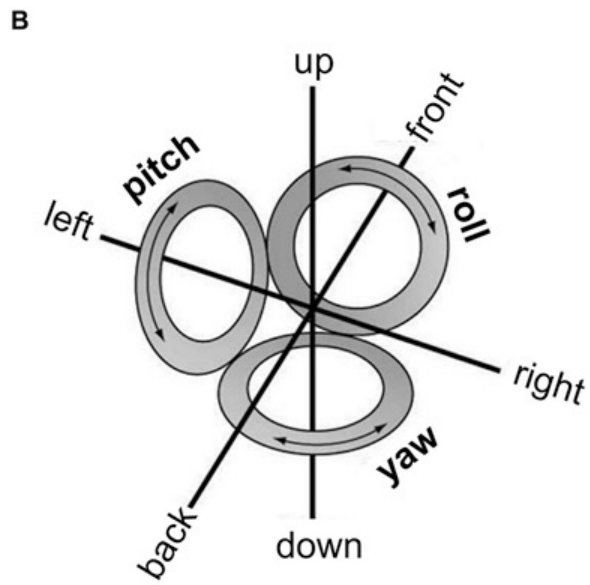
- Simple filtering: Frequency segregation
  - Low frequency motions interpreted as tilt
  - High frequency motions interpreted as translation

- Multi-sensory integration via internal models
  - Proprioceptive and tactile cues
  - Visual information
  - Information from the **semicircular canals**

# SEMICIRCULAR CANALS

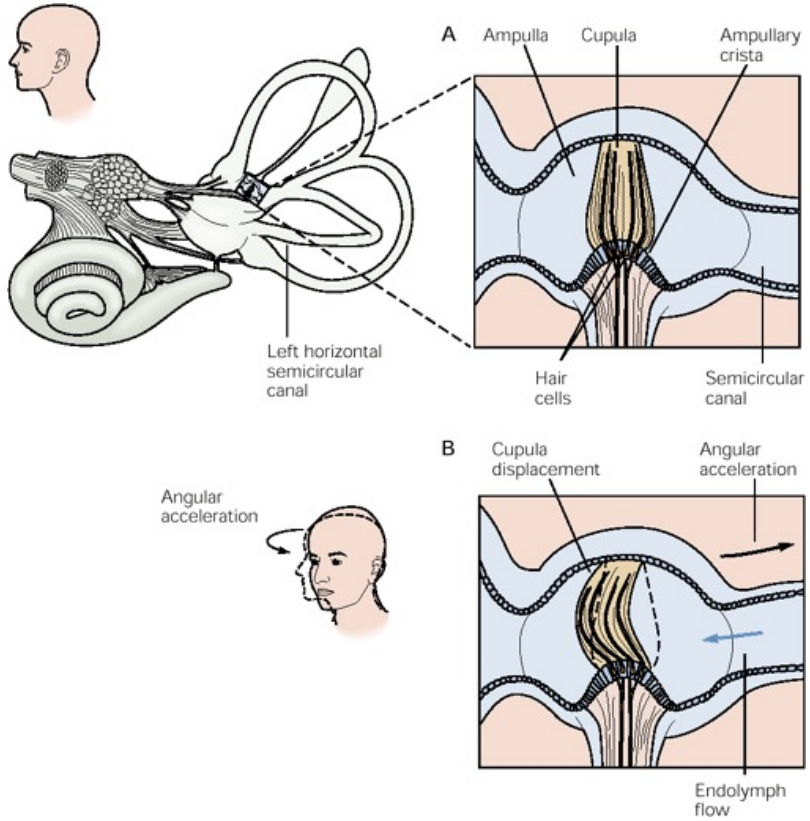



Blausen.com staff (2014)



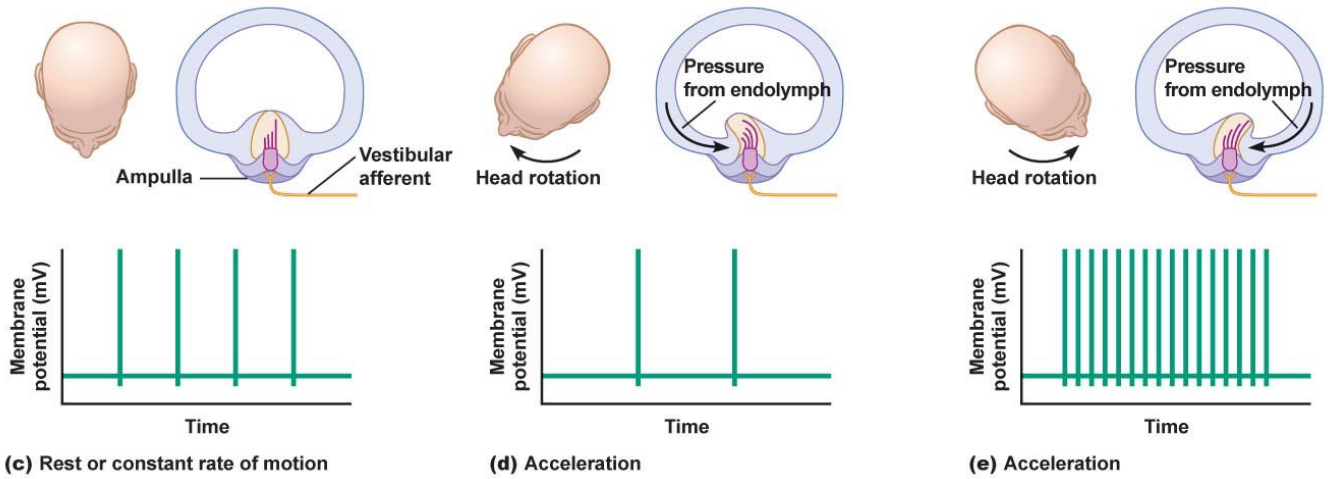

Pfeiffer et al., 2014

# SEMICIRCULAR CANALS



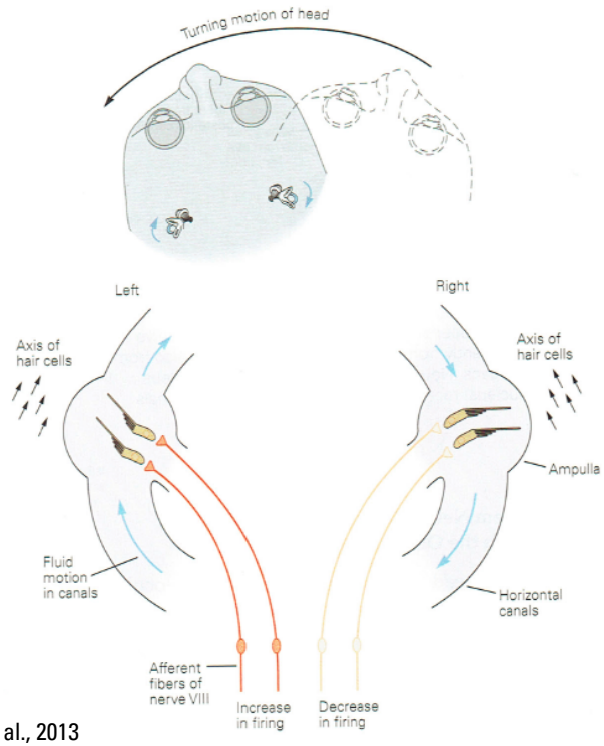

Kandel et al., 2013

# DETECTING ROTATION DIRECTION

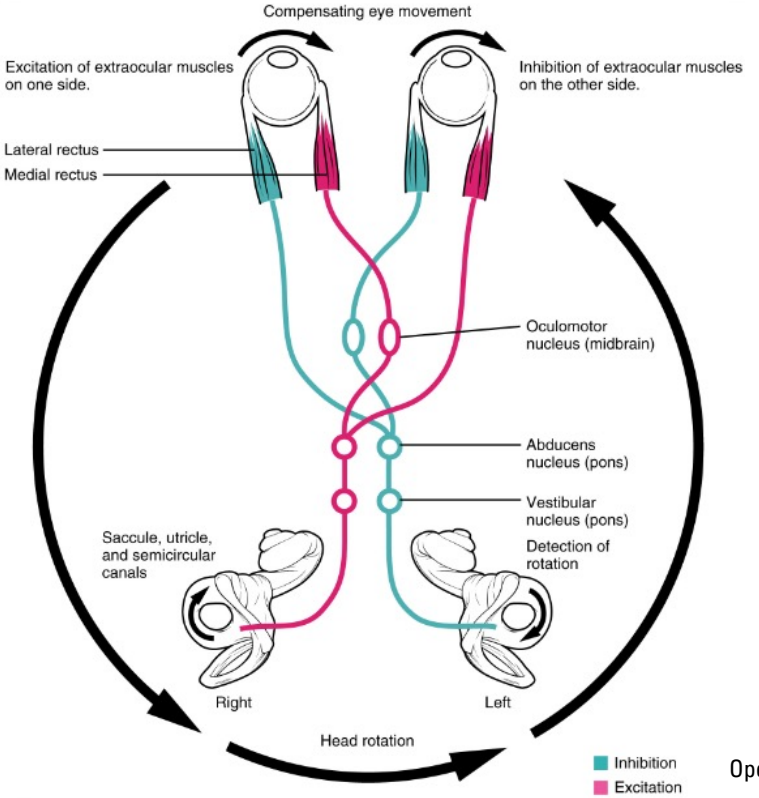



Stanfield, 2011

# DETECTING ROTATION DIRECTION



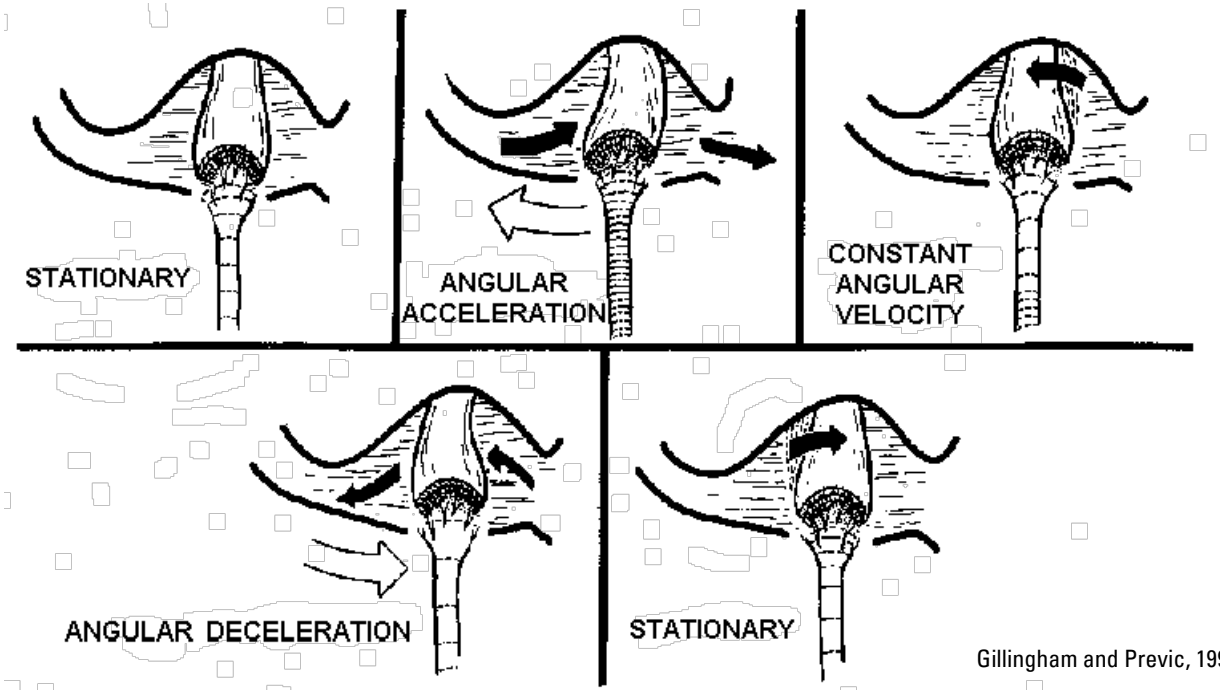

Kandel et al., 2013




Vestibulo-ocular reflex (VOR)

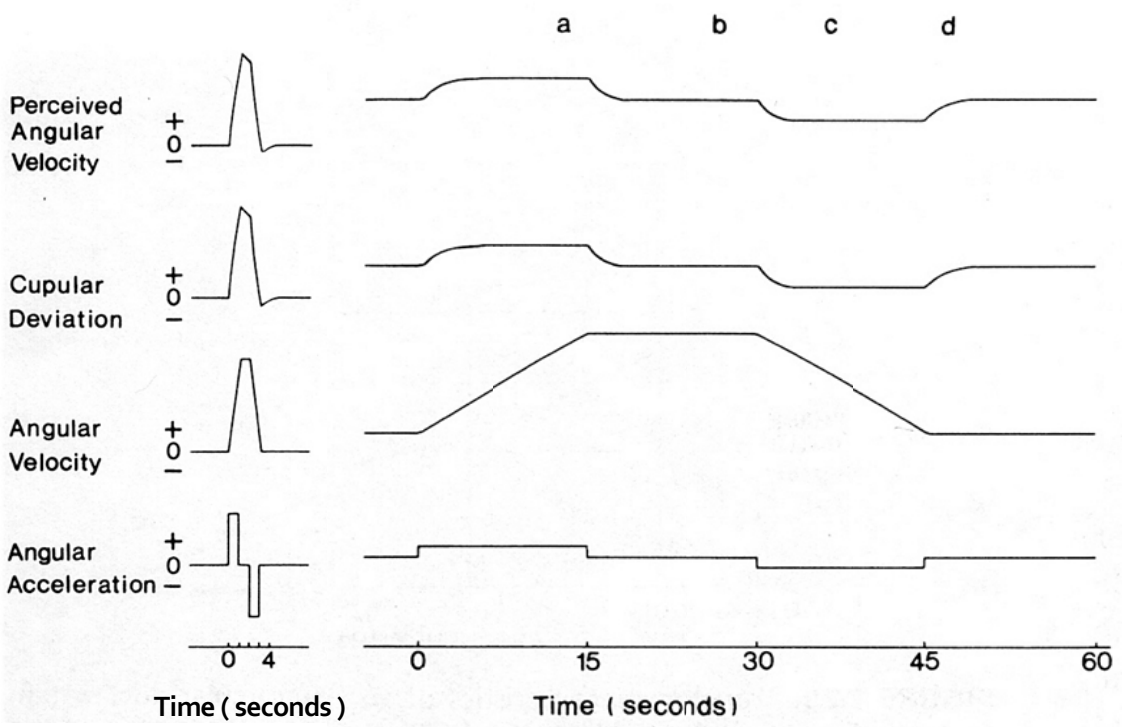
OpenStax College

# CUPULA DEFLECTION




Gillingham and Previc, 1993.

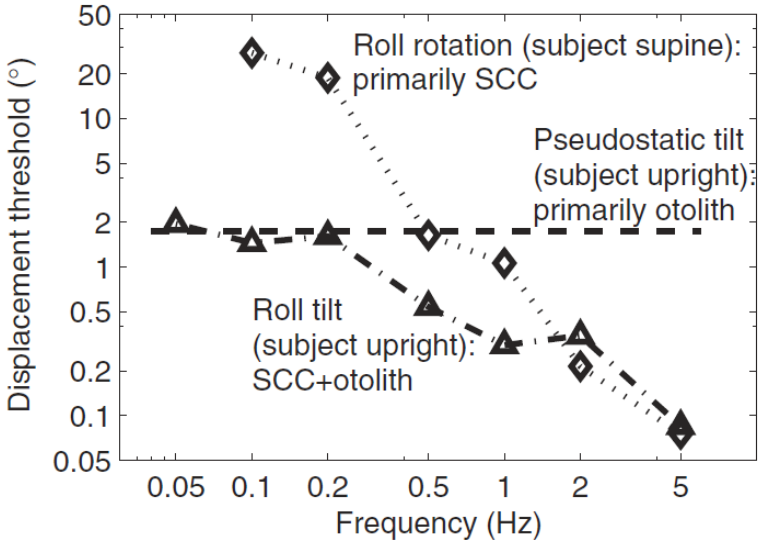
# CANAL RESPONSE DURING AND FOLLOWING A SUSTAINED TURN




Gillingham and Previc, 1993.

# COMBINING SEMICIRCULAR CANAL & OTOLITH SIGNALS

Jenks Vestibular Physiology Laboratory
Massachusetts Eye and Ear Infirmary



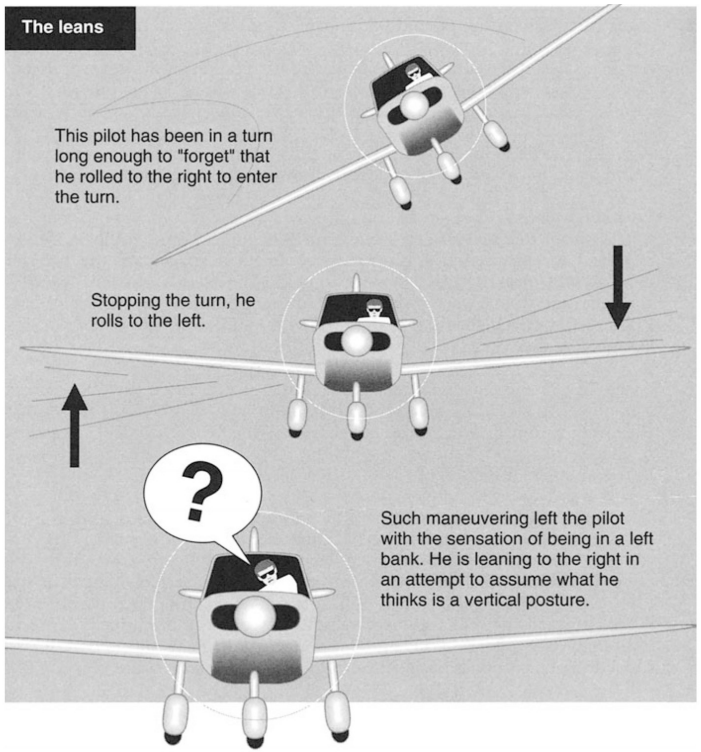

Karmali et al., 2014

# SPATIAL DISORIENTATION

Result from the **normal** sensory processing of signals in an **abnormal** environment

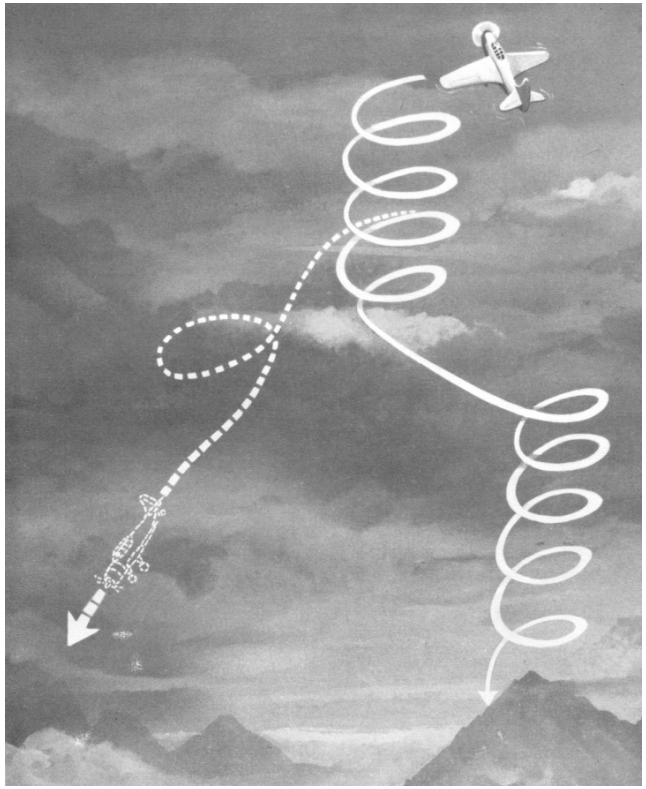
- *Somatogyral Illusions*: False sense of body rotation
- *Somatogravic Illusions*: False sense of body tilt
- *Visual Illusions*: False sense of visual cues

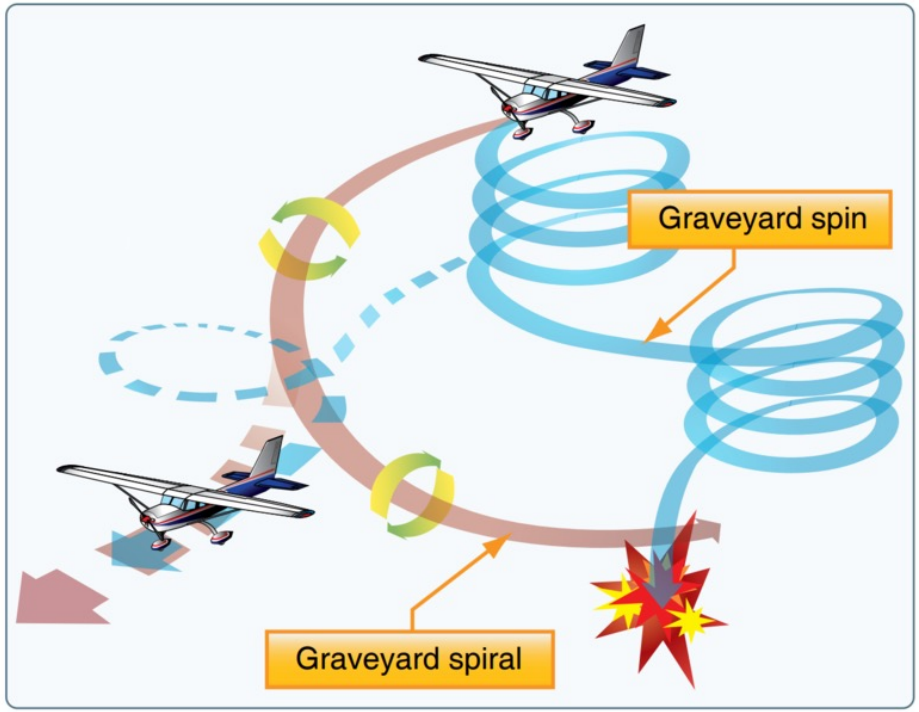
# SOMATOGYRAL ILLUSION: THE LEANS



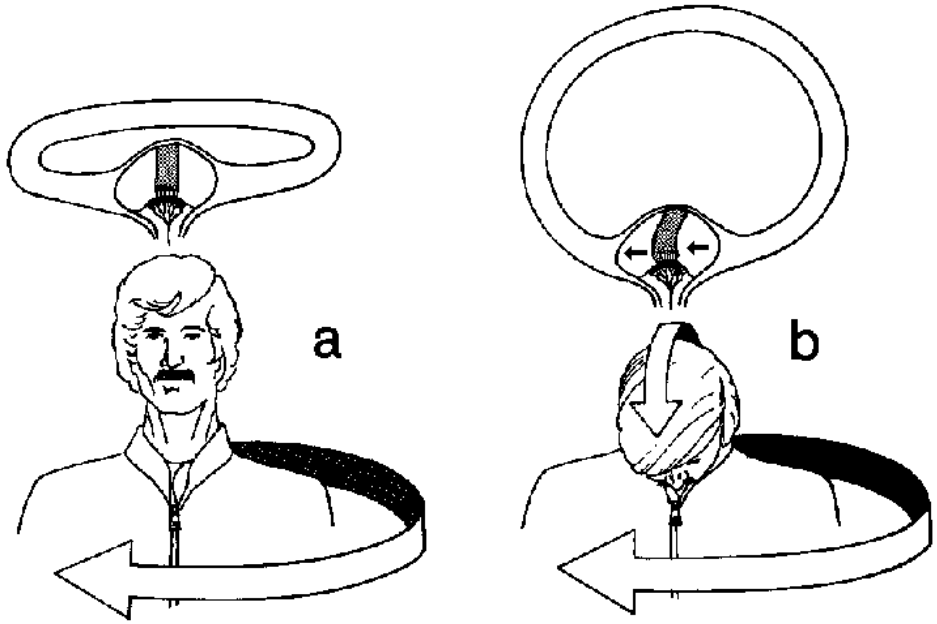

Reinhart, 2008

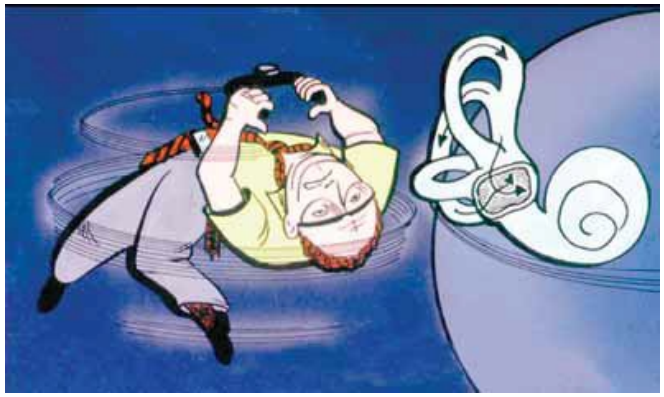
# SOMATOGYRAL ILLUSION: GRAVEYARD SPIN/SPIRAL





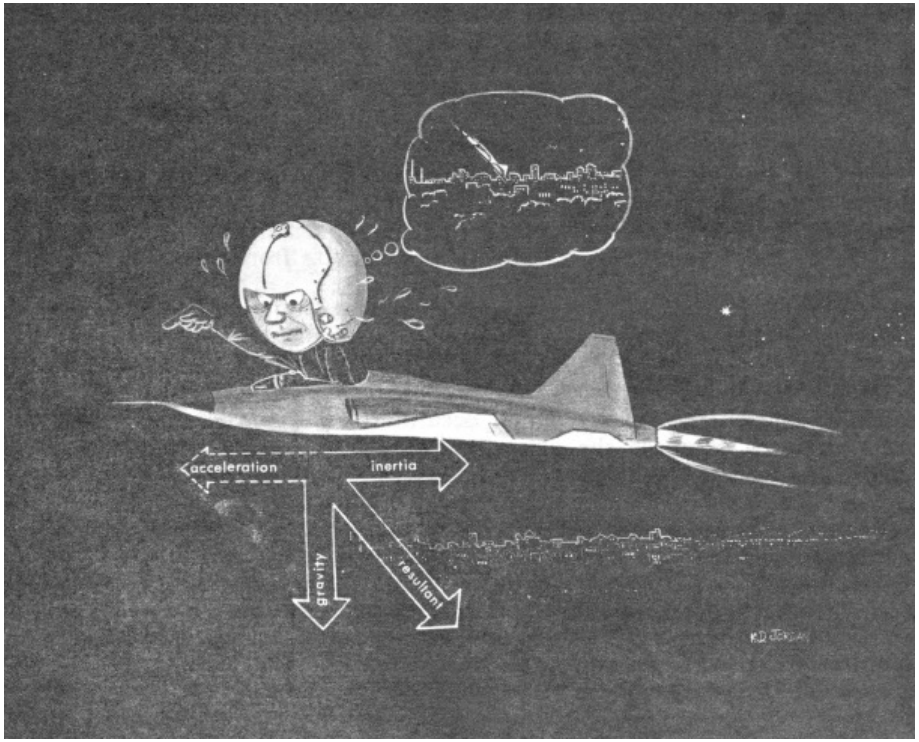
# SOMATOGYRAL ILLUSION—CORIOLIS CROSS COUPLING



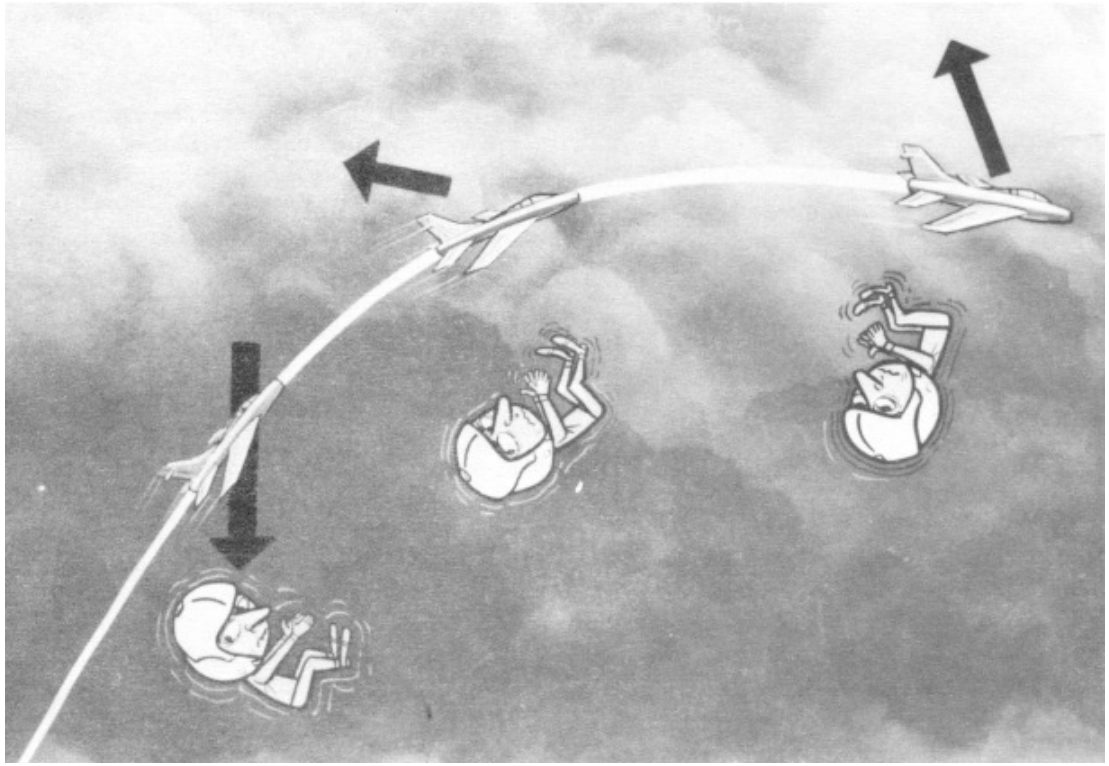
Gillingham and Previc, 1993

# SOMATOGRAVIC ILLUSION—PITCH WHILE ACCELERATING



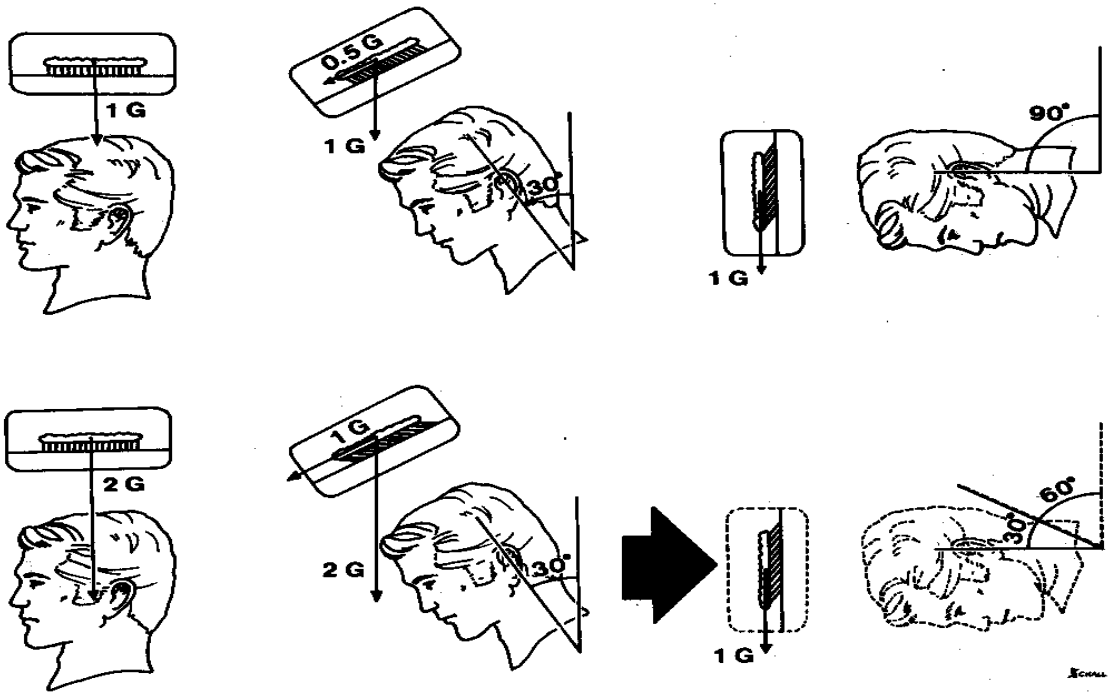

Gillingham and Previc, 1993

# SOMATOGRAVIC ILLUSION—TOP OF CLIMB INVERSION



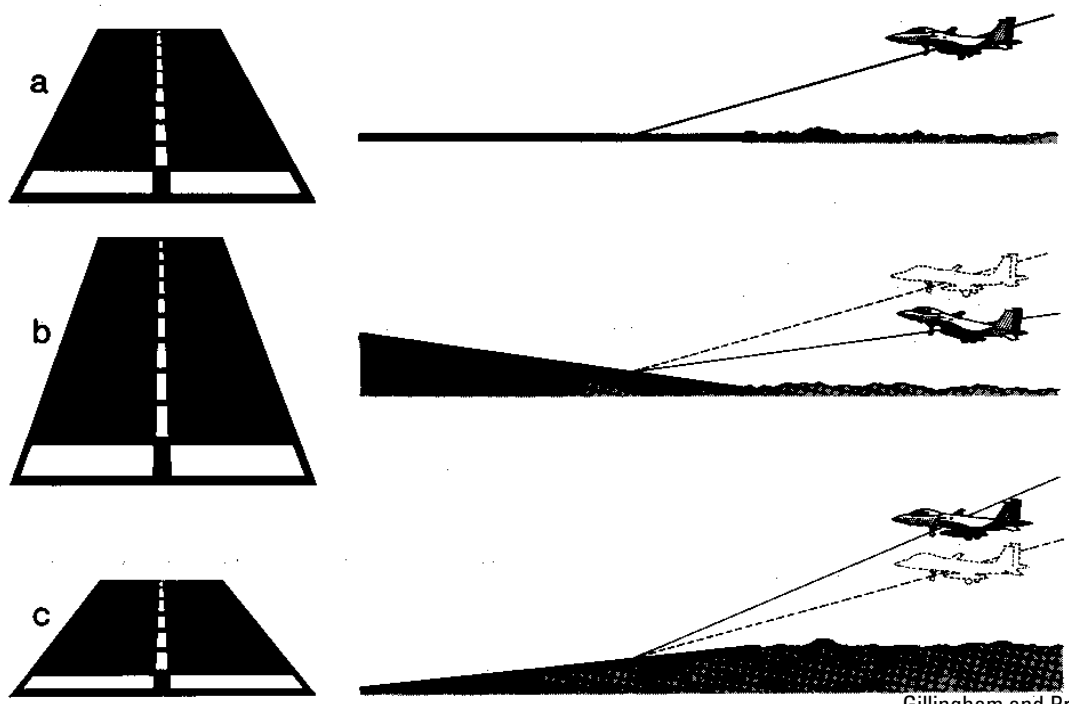
Gillingham and Previc, 1993

# SOMATOGRAVIC ILLUSION—G EXCESS PHENOMENA



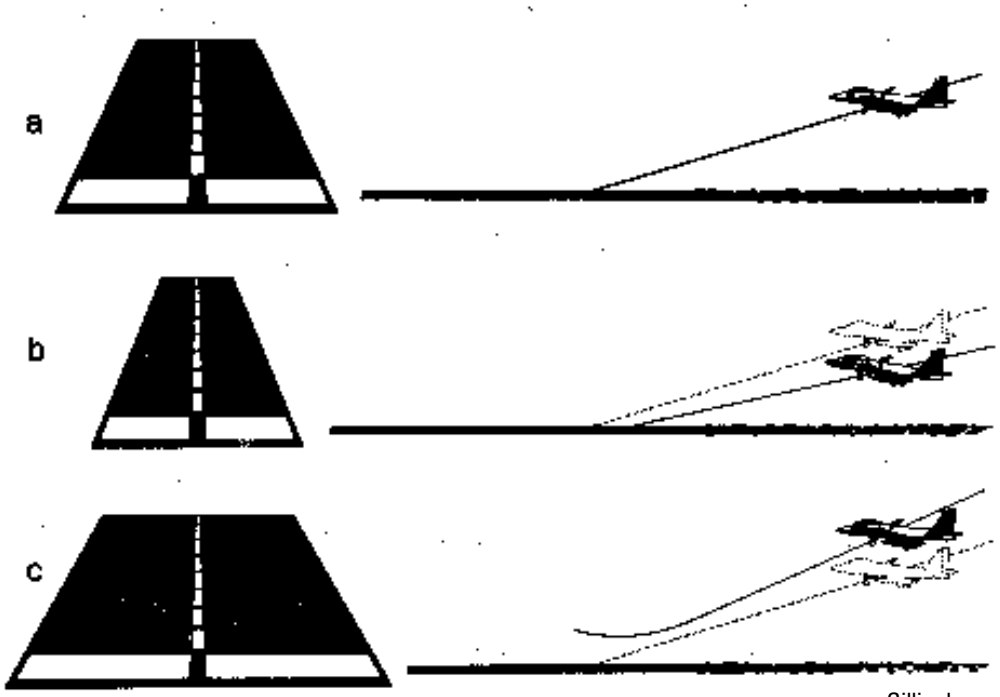

Gillingham and Previc, 1993

# VISUAL FIELD ILLUSION—SHAPE CONSTANCY



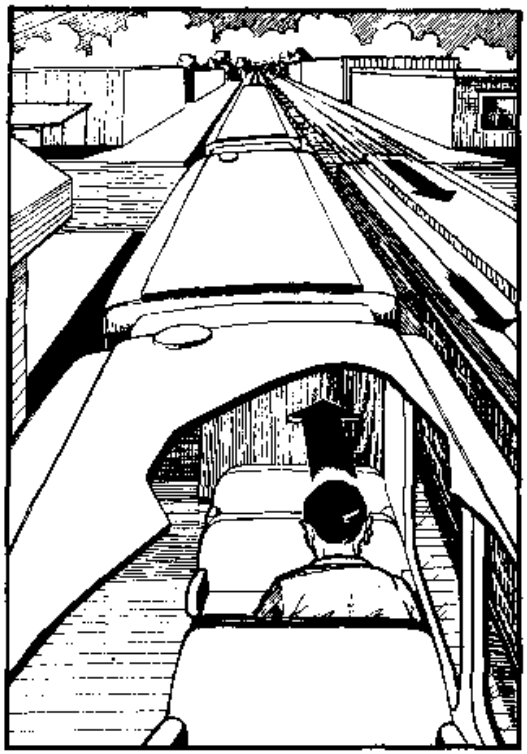

Gillingham and Previc, 1993

# VISUAL FIELD ILLUSION—SIZE CONSTANCY

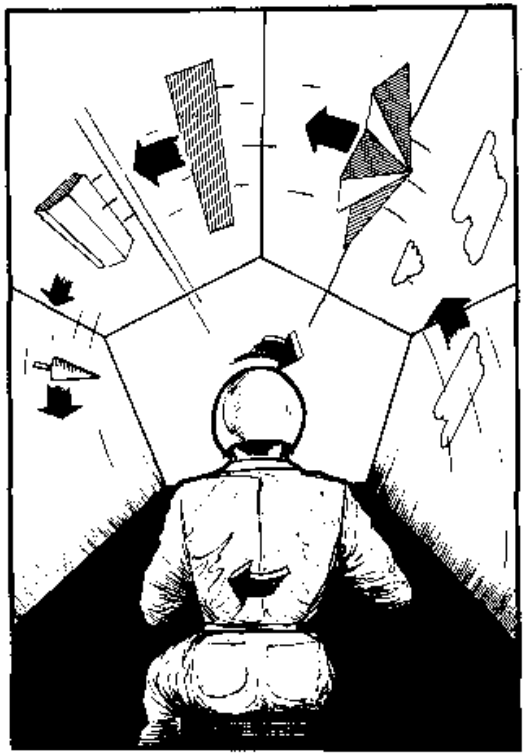



Gillingham and Previc, 1986

# VISUAL FIELD ILLUSION—VECTION



a



b

Gillingham and Previc, 1986

# AVIATION SPATIAL DISORIENTATION – MAJOR CAUSES

- Night Flying
- False Horizons
- Formation Flight
- Transitions to/from looking at instruments
- Head Movements during turns
- High Workload

# SPATIAL DISORIENTATION IN THE DEPT OF DEFENSE

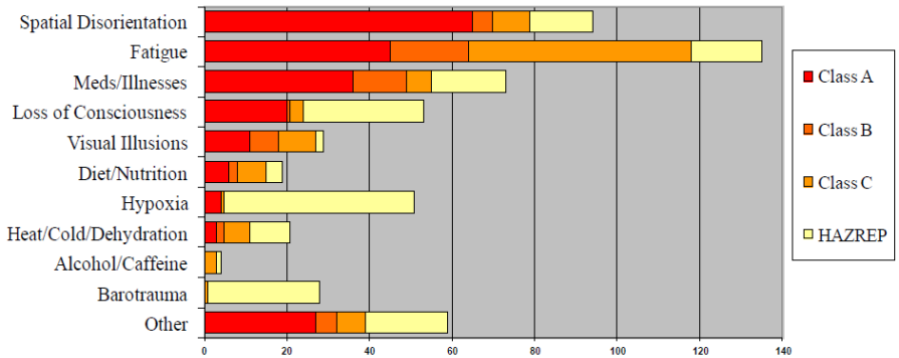
*Spatial disorientation is the leading cause of catastrophic flight mishaps in the DoD (Class A mishaps include fatality or loss of aircraft)*

Air Force Causal Factors for Class A Mishaps 1993-2013

| | Total Mishaps | Fatal Mishaps (N) | Fatal Mishaps (%) | Lives Lost | Aircraft Lost | Cost ($B) |
|---|---|---|---|---|---|---|
| SD | 72 | 44 | 61.1% | 101 | 65 | 2.32 |
| Total | 601 | 129 | 21.5% | 406 | 368 | 13.04 |
| Percent | 12.0% | 34.1% | 34.1% | 24.9% | 17.7% | 17.8% |

Poisson and Miller, 2014

Navy AeroMedical Causal Factors 1990-2011




Belland, 2012

# MINIMIZING AVIATION SPATIAL DISORIENTATION

- Training with exposure to illusions
- Confidence and competence with instrument flying
- Don’t trust vestibular system in ‘abnormal’ environments, especially when low visibility
- Make decisions based on instruments

Cessna 172 Instrument Panel

# MOTION SICKNESS

- Sensory Conflict Theory
  - Motion sickness arises from conflicting cues between sensory channels and expected/experienced stimuli
  - Vision, vestibular system, and proprioceptive system may present conflicting information

http://www.oculusvr.com/rift/

wikicommons

# MOTION SICKNESS SUSCEPTIBILITY

- Signs/symptoms of motion sickness have different functional significance for different individuals
- Susceptibility to one type of provocative stimulation does not necessarily mean susceptibility to another form of stimulation
- Not typically similar susceptibility of motion sickness even for comparable forms of stimulation
  - Example: Motion sickness on Earth does not seem to correlate to **Space** motion sickness.

Lackner and DiZio, 2006

# MOTION SICKNESS MEASUREMENTS

- Susceptibility

MSSQ-Short

| | Not Applicable - Never Travelled | Never Felt Sick | Rarely Felt Sick | Sometimes Felt Sick | Frequently Felt Sick |
|---|---|---|---|---|---|
| Cars | | | | | |
| Buses or Coaches | | | | | |
| Trains | | | | | |
| Aircraft | | | | | |
| Small Boats | | | | | |
| Ships, e.g. Channel Ferries | | | | | |
| Swings in playgrounds | | | | | |
| Roundabouts in playgrounds | | | | | |
| Big Dippers, Funfair Rides | | | | | |
| | t | 0 | 1 | 2 | 3 |

- Symptom severity

In the **last 30 minutes**, how severe is your motion sickness?

Please circle the number below that best represents your response:

*Please select only one response.*

| **0** | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| No symptoms | Stomach awareness or discomfort | Mild nausea | Moderate nausea | Severe nausea | Retching | Vomiting |

# REFERENCES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